CYBER SECURITY CASE STUDY REPORT

Social Engineering Attacks and Major Security Breaches

How Twitter, Target and MGM Resorts were attacked, what it cost, and how they recovered

Submitted by	Karmanya joshi
Enrollment No.	241105060065
Course / Subject	Diploma it
Faculty	Prof. Tarjanee Vyas
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University	GMIU

Overview

Social engineering attacks target people rather than software. The attacker builds trust or urgency and persuades someone to hand over access. This report reviews three well-known breaches, each driven by a different technique, and compares how the companies were hit, what the damage was, and how they recovered.

Company	Year	Technique	Main outcome
Twitter	2020	Phone spear phishing, pretexting	130 accounts targeted, Bitcoin scam, internal tools locked down
Target	2013	Spear phishing of a vendor	About 40 million cards and up to 70 million customer records exposed, about $292 million cost
MGM Resorts	2023	Vishing, help desk pretexting	About $100 million impact, ten-day disruption, customer data stolen

Case Study 1: Twitter (2020)

Technique: phone-based spear phishing and pretexting of employees

1. Company Description

Twitter (now X) was a global social media platform founded in 2006 and headquartered in San Francisco, California. It lets users publish short public messages and send direct messages. In mid-2020 it had well over 180 million daily users, including heads of state, celebrities, and major companies whose verified accounts carried real influence.

Industry	Social media / technology
Headquarters	San Francisco, California, USA
Date of attack	15 July 2020
Attack type	Social engineering: phone spear phishing (vishing) and pretexting

2. How the Attack Happened

The attackers did not break Twitter's software. They studied which employees had access to internal support tools, then phoned them while pretending to be internal colleagues. Twitter described it as a phone spear-phishing attack that exploited human vulnerabilities.

- **Research:** Attackers identified a small group of employees with access to account support tools.
- **Deception:** They called these employees, posed as trusted staff, and talked them into giving up login credentials.
- **Access:** With valid credentials, they entered Twitter's internal administration tools.
- **Exploitation:** They took control of 130 targeted accounts and posted a Bitcoin giveaway scam from 45 of them, including accounts of well-known public figures.

3. Article Photo



Figure 1: *[Add source: article title, publisher, date]*

4. Steps Taken to Recover

Twitter's response focused on containing the damage quickly and then limiting who can reach powerful internal tools.

- **Contained the incident:** Locked the affected accounts, removed the scam tweets, and temporarily restricted tweeting from many verified accounts.
- **Reduced internal access:** Significantly limited access to internal tools and systems, which also slowed some support requests for a while.
- **Worked with law enforcement:** Cooperated with an investigation that led to arrests within weeks of the attack.
- **Improved security and training:** Announced additional security steps and stronger employee awareness against phone-based social engineering.

5. Cost and Data Lost

Area	Details
Direct financial loss	Twitter did not publish a total cost. The scam itself collected roughly $113,500 in Bitcoin from victims, according to press reports.
Accounts affected	130 accounts targeted; tweets sent from 45.
Data accessed	Direct message inboxes viewed for 36 accounts; account data downloaded for 7 accounts.
Other costs	Investigation, legal and regulatory attention, and reputational damage (no official figure disclosed).

6. Critical Impacts

- **Trust:** Verified accounts of public figures were hijacked, which raised doubts about platform security.
- **Security of a single point:** A handful of employees' credentials gave access to a very powerful toolset.
- **Scrutiny:** State regulators and law-enforcement agencies examined how Twitter protected internal systems.

7. Problems and Solutions

Problem (what went wrong)	Solution (what fixes it)
Employees could be convinced by a phone caller to share credentials.	Phishing-resistant multi-factor authentication (security keys), plus training on voice scams and call-back verification.
Too many staff could use powerful account-control tools.	Least-privilege access, approval for sensitive actions, and detailed logging.

Problem (what went wrong)	Solution (what fixes it)
Attack was detected only after tweets went out.	Monitoring for unusual admin activity and rapid account lock-down procedures.

Case Study 2: Target Corporation (2013)

Technique: spear phishing of a third-party vendor

1. Company Description

Target Corporation is one of the largest retailers in the United States, headquartered in Minneapolis, Minnesota. It operates general-merchandise stores and an online business, and processes millions of card payments each week, which makes payment data its most sensitive asset.

Industry	Retail
Headquarters	Minneapolis, Minnesota, USA
Period of attack	November to December 2013 (peak holiday shopping season)
Attack type	Spear phishing of a vendor, then malware on point-of-sale systems

2. How the Attack Happened

Attackers did not go after Target directly. They sent a spear-phishing email to Fazio Mechanical Services, a small heating and cooling (HVAC) contractor that had limited network access to Target, and stole its credentials.

- **Phishing the vendor:** A targeted email to the HVAC contractor led to stolen vendor login credentials.
- **Entering the network:** The credentials gave access to Target's vendor systems, and the attackers then moved deeper into the internal network.
- **Installing malware:** Malware was placed on point-of-sale registers and collected card data as customers swiped their cards.
- **Data theft:** Stolen data was sent out of the network over several weeks before it was discovered.

3. Article Photo



Figure 2: *[Add source: article title, publisher, date]*

4. Steps Taken to Recover

Target's recovery combined technical fixes, leadership changes, and a heavy investment in payment security.

- **Investigation and containment:** Engaged outside forensic experts, removed the malware, and cooperated with law enforcement and payment networks.
- **Customer support:** Offered free credit monitoring to affected customers.
- **Payment security:** Accelerated the move to chip-and-PIN card technology in its stores.
- **Security organisation:** Reported leadership changes, including the departure of the CEO, and strengthened its information security team and monitoring.
- **Legal settlements:** Settled with card networks, banks and credit unions, consumers and U.S. states.

5. Cost and Data Lost

Area	Details
Total reported cost	About $292 million in breach-related expenses, or about $202 million after insurance reimbursements, as reported in Target's financial disclosures.
Card data stolen	About 40 million credit and debit card records.
Personal data stolen	Names, addresses, phone numbers and email addresses of up to 70 million customers.
Settlements	$19 million (Mastercard), $67 million (Visa), $39.4 million (banks and credit unions), $10 million (consumer class action), and $18.5 million (multistate agreement).

6. Critical Impacts

- **Customer trust:** The breach happened days before Christmas and drew national headlines.
- **Leadership:** The CEO stepped down in the aftermath.
- **Third-party risk:** Showed that a small supplier can become the weakest link into a large company.
- **Regulation:** Triggered Senate attention and a settlement that required stronger security practices.

7. Problems and Solutions

Problem (what went wrong)	Solution (what fixes it)
A vendor's credentials were stolen through phishing.	Multi-factor authentication for vendors and security awareness requirements in supplier contracts.

Problem (what went wrong)	Solution (what fixes it)
Vendor access reached sensitive internal systems.	Network segmentation so vendor accounts cannot touch payment systems.
Malware ran on registers for weeks undetected.	Alert monitoring, application allow-listing on point-of-sale devices, and chip-and-PIN cards.

Case Study 3: MGM Resorts International (2023)

Technique: voice phishing (vishing) and pretexting against the IT help desk

1. Company Description

MGM Resorts International is a global hospitality and entertainment company based in Las Vegas, Nevada, operating since 1986. It runs large casino resorts such as Bellagio, MGM Grand and The Mirage, where hotel check-in, room keys, slot machines and payments all depend on connected IT systems.

Industry	Hospitality, gaming and entertainment
Headquarters	Las Vegas, Nevada, USA
Date of attack	About 9–11 September 2023
Attack type	Vishing and pretexting of the help desk, followed by ransomware

2. How the Attack Happened

According to the group that claimed responsibility (Scattered Spider) and later US government advisories on its methods, the attackers found an employee's details on LinkedIn, phoned the IT help desk pretending to be that person, and persuaded staff to reset the credentials and multi-factor authentication. The same group worked with the ALPHV/BlackCat ransomware gang.

- **Research:** Employee information was gathered from public sources such as LinkedIn.
- **Impersonation:** A phone call to the help desk led to a password and MFA reset for the impersonated account.
- **Privilege escalation:** The attackers moved through MGM's identity systems and gained high-level access.
- **Ransomware:** Over a hundred virtual server hosts were encrypted, and the attackers also claimed to have stolen customer data.

3. Article Photo



Figure 3: *[Add source: article title, publisher, date]*

4. Steps Taken to Recover

MGM took systems offline itself to stop the spread, then rebuilt service by service.

- **Shut down systems:** Took affected systems offline proactively and brought in outside cybersecurity experts and law enforcement.
- **Manual operations:** Hotels and casinos kept running with manual processes while digital services were down.
- **Restoration:** Systems were restored over roughly ten days. Reports say MGM did not pay the ransom.
- **Customer protection:** Notified affected customers and offered free credit monitoring and identity protection.
- **Financial protection:** Expected most of the loss to be covered by cyber insurance.

5. Cost and Data Lost

Area	Details
Financial impact	About $100 million negative impact on quarterly earnings, plus less than $10 million in one-time costs for remediation, legal fees, advisers and incident response (SEC filing, October 2023).
Data stolen	Personal details of some customers who transacted before March 2019, including names and contact details, and for a smaller group more sensitive identifiers.
Operations disrupted	Slot machines, digital room keys, check-in, payments and reservations were affected for about ten days.

Area	Details
Attackers' claim	The group claimed to have taken about 6 TB of data (an attacker claim, not confirmed by MGM).

6. Critical Impacts

- **Business interruption:** Core revenue systems were down at peak hotel and casino operations.
- **Customer data exposure:** Identity-theft risk for affected guests.
- **Help desk risk:** A single phone call was enough to start a company-wide incident.
- **Legal exposure:** Regulators and customers raised legal action after the incident.

7. Problems and Solutions

Problem (what went wrong)	Solution (what fixes it)
Help desk reset credentials without strong identity proof.	Strict verification (manager approval, video check, call-back to a known number) for password and MFA resets.
One account could reach powerful identity systems.	Least privilege, separate admin accounts, and extra approval for privileged resets.
Ransomware spread widely once inside.	Network segmentation, offline backups, and a tested incident-response plan.

Comparison and Key Lessons

	Twitter (2020)	Target (2013)	MGM (2023)
Entry point	Employees by phone	Third-party vendor by email	IT help desk by phone
Data or assets hit	Accounts, DMs	Card and customer data	Customer data, all systems
Reported cost	Not disclosed	About $292 million	About $100 million plus under $10 million
Core weakness	Over-broad internal access	Vendor access and weak segmentation	Weak identity checks for resets

- **People are the target:** In all three cases the first failure was a human decision, not a software flaw.
- **Verify through a second channel:** Never trust a caller or email on its own, especially when it asks for credentials or resets.
- **Limit access:** Least privilege and segmentation turn a stolen login from a disaster into an incident.
- **Use strong authentication:** Security keys and approval workflows resist phishing better than passwords and SMS codes.
- **Prepare to respond:** Tested incident plans, offline backups and insurance shorten recovery.

References

- Netwrix. MGM cyber attack analysis. netwrix.com/en/resources/blog/mgm-cyber-attack/
- BleepingComputer. MGM Resorts says ransomware attack cost $100 million, data stolen. bleepingcomputer.com
- The SSL Store. Cost of the 2013 Target data breach nears $300 million. thesslstore.com
- Sophos. Target settles with banks for $39 million. news.sophos.com
- Simpson Thacher & Bartlett. Target consumer class action memo (April 2015). stblaw.com
- Associated Press. Twitter says hackers used phone to fool staff, gain access (July 2020).
- CFO.com. Twitter Bitcoin hack caused by phone-based phishing attack. cfo.com
- [ADD THE ARTICLE LINKS YOU USED FOR THE PHOTOS]